quasiparticle spectrum, and thus the response of such supercondcutors to microwave and acoustic fields. Indeed recent experiments using immersion cooling of superconducting qubits and resonators in liquid $^3$He show that the TLS population is relaxed substantially to their ground levels [41]. In particular, the microwave power required to saturate the TLS levels increases by three orders of magnitude with immersion cooling, suggesting that the TLS populations may be substantially out of equilibrium for superconducting devices operating under microwave or current excitation.

To exhibit the sensitivity of the spectrum of sub-gap quasiparticle states to the TLS level population start from the specral function of the TLS impurity self energy,

$$\begin{aligned}\mathrm{Im}\,\widehat{\Sigma}^{\mathrm{R}}(\varepsilon) = \frac{\hbar}{2\pi\tau_{in}}\int dE\,p(E)\,\Big\{\;&\mathrm{Im}\,\widehat{\mathfrak{G}}^{\mathrm{R}}(\varepsilon-E)\,\Big[\,N_g(E)-N_{ge}(E)\;n_{\mathrm{F}}(\varepsilon-E)\Big]\\ &+\;\mathrm{Im}\,\widehat{\mathfrak{G}}^{\mathrm{R}}(\varepsilon+E)\,\Big[\,N_e(E)+N_{ge}(E)\;n_{\mathrm{F}}(\varepsilon+E)\Big]\Big\}\,,\end{aligned} \tag{73}$$

where the dependence on the occupation of the ground and excited levels is highlighted. Multiple factors can lead to TLS levels being out of equilibrium with the bath of electrons and phonons, including the relaxation timescales of the TLS levels via electron and phonon interactions, microwave fields coupling to TLS impurities, and screening currents driving the TLS distributions. Here we model the out of equilibrium TLS level population by an effective temperature of the TLS level population, $T < T_{tls} \leq 2.5\,T_{c_0}$. The nonequilibrium TLS population dramatically changes the spectrum of sub-gap states, particularly near the Fermi energy ($\varepsilon = 0$), providing a mechanism for pair-breaking that spans all energies below the clean limit excitation gap, in a way that is remarkably similar to the ad hoc Dynes pair-breaking parameter [42].

Figure 9 shows the impact of the nonequilibrium level populaton of TLS impurities parametrized by the effective TLS bath temperature on the quasiparticle density of states for four TLS level distributions, and with electrons and phonons at $T = 0.1 - 0.2T_{c0}$. The primary observation is that the sub-gap density of states at the Fermi level is exponentially sensitive to the TLS level population that is out of equilibrium. The quasiparticle DOS is also sensitive to the distribution of TLS energy level splittings. In particular, scattering off tunneling impurities with distributions that have a substantial density of high-energy level splittings generate the largest DOS at the Fermi level. Thus, the glass-like distribution with $E_c = 3.0\,T_{c_0}$ exhibits the largest DOS at the Fermi level, while the modified Wipf-Neumaier distribution, which is dominated by TLS impurities with much lower-energy tunnel splittings, exhibits a finite, but significantly lower DOS at the Fermi level.

The significance of the sub-gap quasiparticle spectrum generated by out of equilibrium TLS impurities is the *new channel for microwave power loss* from quasiparticle dissipation that results from pair-breaking by scattering from non-equilibrium TLS impurities. A quantitative analysis of quasiparticle dissipation generated by the non-equilibrium TLS population is the subject of a separate study that includes the population dynamics of the TLS impurity distribution driven by a microwave field.

## VII. RESUMÈ

We have developed a quantum field theory formulation of superconductors interacting with a random distribution of TLS impurities. The basic interaction is formulated as a scattering theory of quasiparticles, including those bound as Cooper pairs, and TLS impurities, with the internal state of each impurity described by an $\mathsf{SU}(2)$ isospin. Our formulation represents each TLS isospin by a local fermion field à la Abrikosov [26], but projects out unphysical states based on Popov and Fedetov's method [27] that ensures the resulting perturbation theory in the quasiparticle-TLS interaction obeys the Wick theorem. The most important scattering processes are those in which incoming and outgoing quasiparticles exchange energy as a TLS impurity undergoes a transition between its ground and excited state. This inelastic process leads to both pair-enhancement and pair-breaking.

Pair-enhancement dominates at low temperatures under equilibrium conditions of quasiparticles, phonons and TLS impurities, and leads to modest increase of $T_c$ and the superconducting order parameter, $\Delta$. Pair breaking is evident from sub-gap quasiparticle states, even for TLS impurities in equilibrium with the thermal bath. A key result of our analysis is the sensitivity of the sub-gap density of states to the level populations of the distribution of TLS impurities. Thus, under excitation, or decoupling from the thermal bath, the resulting nonequilibrium level population of the TLS distribution generates subgap quasiparticle states down to the Fermi level which contribute to dissipation and thus degrade the performance of superconducting devices.

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